Sharing a Plot

Whether during our standup or weekly meetings, and even sharing with colleagues and presenting results, this is a good guideline on how to present your plots.

1 Aim

"What am I trying to understand with this plot?"

State in one sentence what you are trying to see.

> *"I'm checking whether my observed galaxy colours are consistent with the expected passive and active templates, or whether there is an intermediate population between the two."*

2 Describe the plot

"What is on each axis, and what are my datasets?"

Be precise, name the axes, lines and datasets first.

> *"x = redshift, y = rest-frame g–r colour. Red = expected passive template, blue = expected active template from simulation XYZ, grey = observed galaxies from my sample after selection cuts. I use error bars from X or derived using this equation."*

3 What do I see?

"What pattern, trend, or anomaly stands out?"

Say what you see, not what you hoped to see. Quantify it. Check whether the feature is significant given the errors.

> *"There is a 0.2 mag colour offset at $z > 0.6$ in the data but not in the simulated templates. At $z > 0.8$ there is a larger dispersion in the colour, with a median of X and a standard deviation of Y. If I take into account statistical errors for that redshift bin as I don't have many samples, this is negligible with respect to the $0.6 > z > 0.8$ interval."*

4 Literature check

"Does this agree with what others have found? By how much?"

Give a number, not just "similar" or "different". If there is a trend to be fitted, do so and quantify it.

> *"This is 2σ larger than the offset in Bruzual et al. 2003 where they predict X."*

5 What didn’t work?

"What did I already try, and why do I think it failed?"

List what you tried so we don't repeat it in the meeting. If you can hypothesise why it didn't work, even better as a discussion point.

> *"A colour cut didn't remove it. Sigma-clipping made it worse. I tried applying a dust correction but it moved the outliers further from the templates, not closer."*

6 Next step

"What is my best hypothesis for what to try next?"

Come with one concrete idea on what to change, or how to validate this.

> *"I want to check whether the blue outliers are X type of galaxies, if so, I may need a third template. I could cross-match with a spectroscopic catalogue to check … I want to split by morphology and see if the offset disappears…"*

AS YOU PROGRESS: ADDITIONAL DETAILS TO INCLUDE

- What can this plot *not* tell you? Be explicit about the limits of this diagnostic.
- Flag any sample size or selection effects that may affect the comparison.
- Note where the code or selection cuts are documented so you, or someone else, can reproduce it in six months.

Follow-up through Slack or other media

When you share a revised plot. Don't just drop an image. Always include enough context so we can provide useful feedback.

What this plot is

One line restating the aim and axes — don’t assume they remember.

"This is the redshift vs g–r colour plot we discussed on [date], checking for an intermediate galaxy population."

What changed

State exactly what you modified.

"I've applied the flux limit cut we discussed and added error bars derived from bootstrap resampling."

What you now see

Did the change have the expected effect? Quantify the difference from the previous version.

"The offset at z > 0.6 reduced from 0.2 to 0.12 mag — still 2σ, so the flux limit was not the main driver."

Next step

"I have been thinking that discrepancy may be due to the size of the catalogue I am using, I will derive the normalisation factors. I'm planning to start from the raw numbers but may cross-check with the paper."

What you need

Be explicit about whether you want feedback, approval to move on, or just a heads-up.

"Happy to proceed to the morphology split — let me know if you see something I'm missing."